

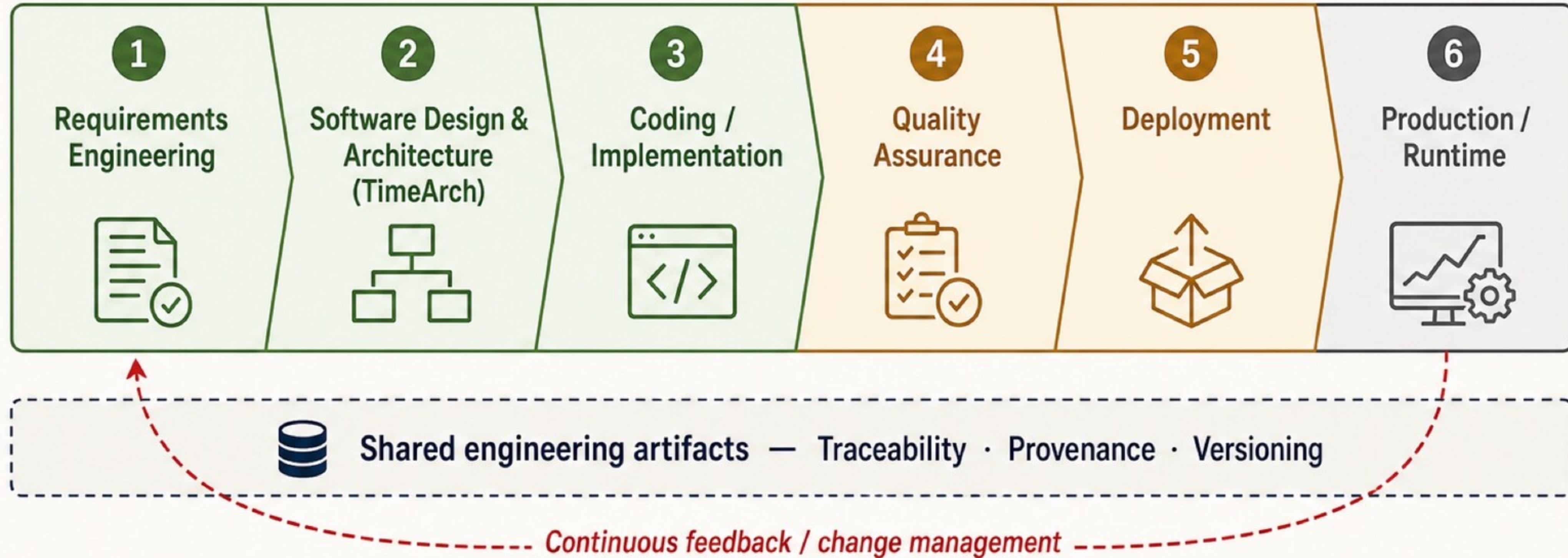
ANSE + KERKIS

Artifact-Driven Autonomous Software Engineering

Process figures for the research wiki

ANSE toolchain — one connected lifecycle

Artifact-Driven Autonomous Software Engineering



KERKIS — Orchestration & Integration

Agent orchestration

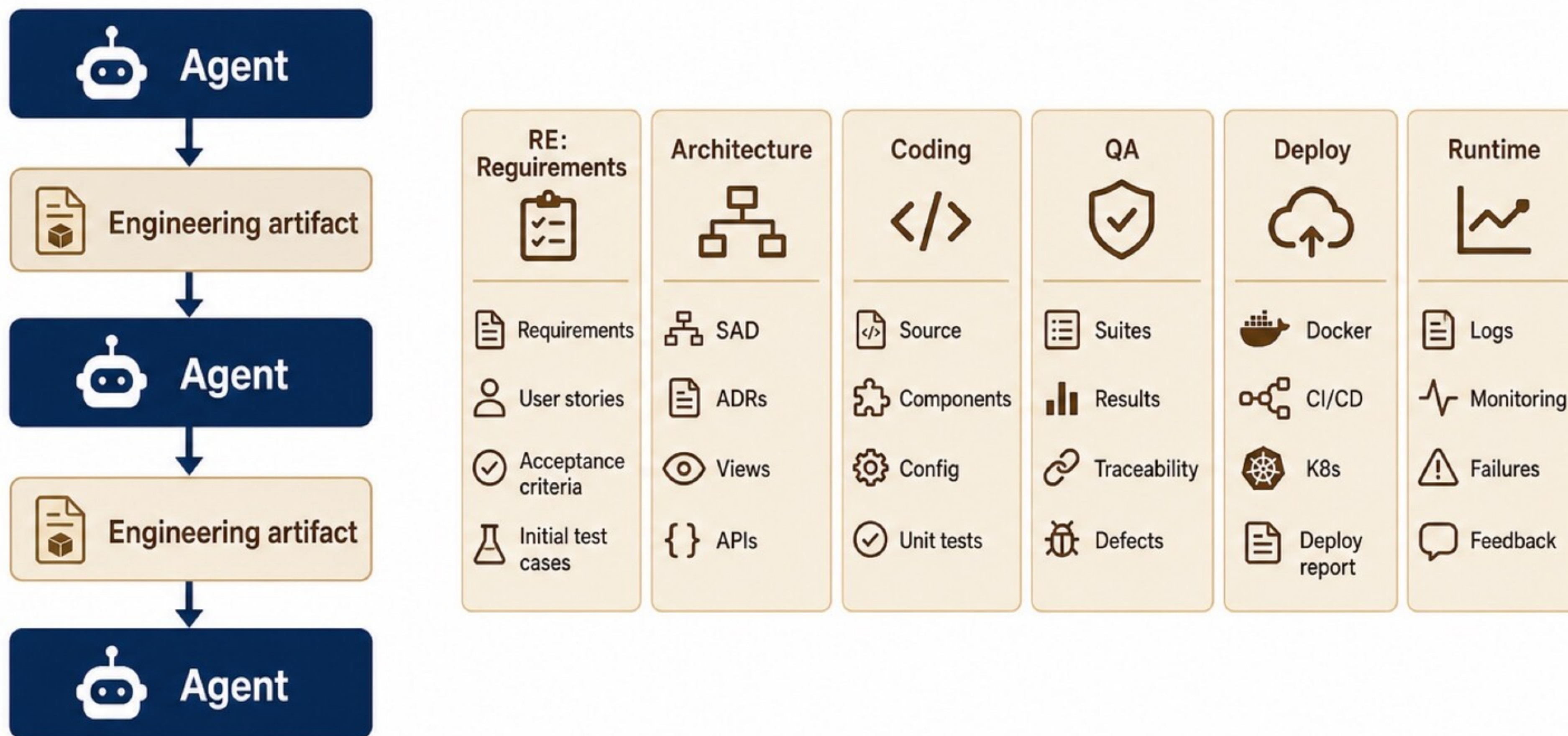
Artifact management

Traceability

Feedback handling

Artifact-driven engineering

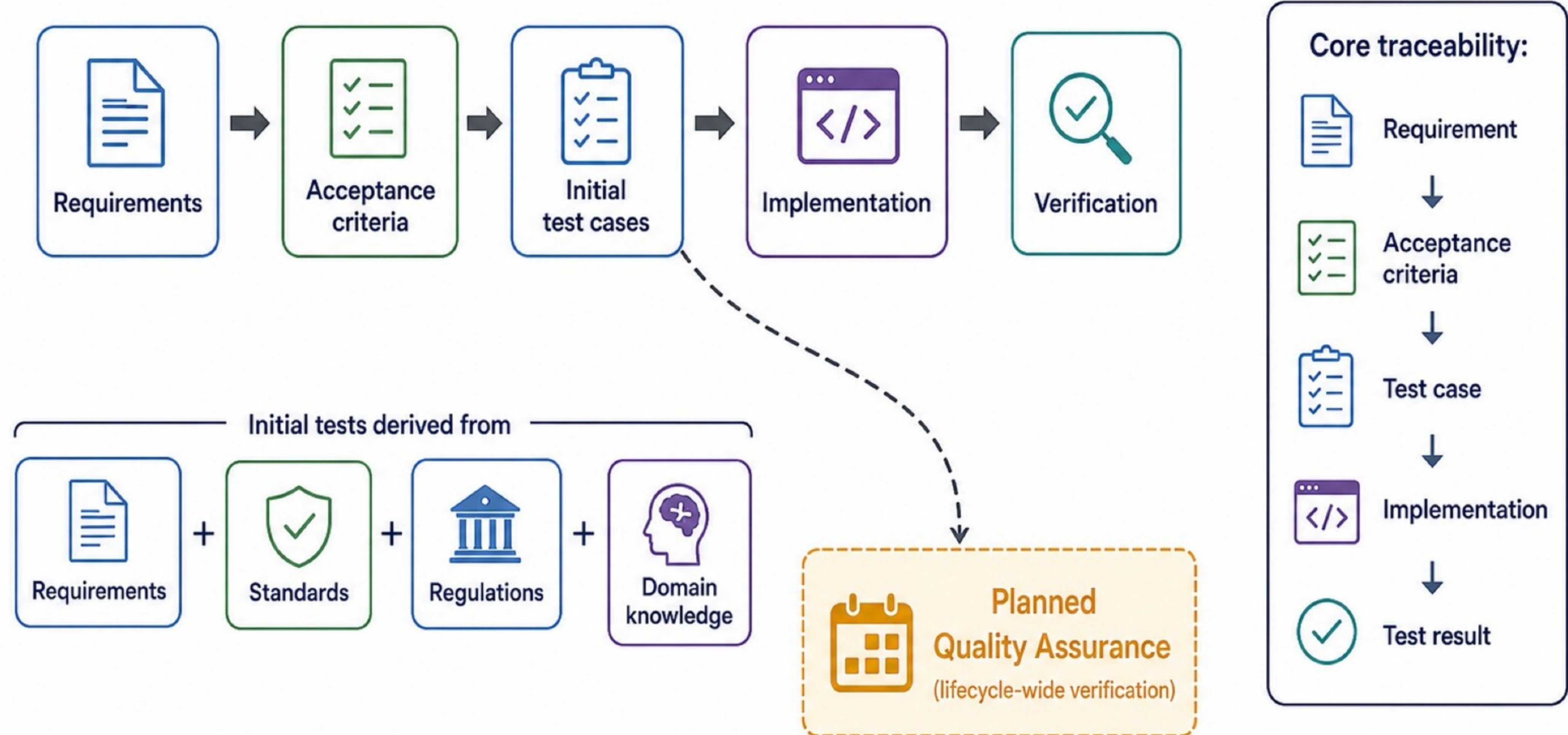
Agents collaborate through explicit, traceable software artifacts rather than only through prompts.



Each artifact carries provenance, version, producer, consumer, relationships.

Requirements Engineering already produces tests

Testing begins early — QA is not independent of RE



Quality Assurance & Deployment — next ANSE capabilities

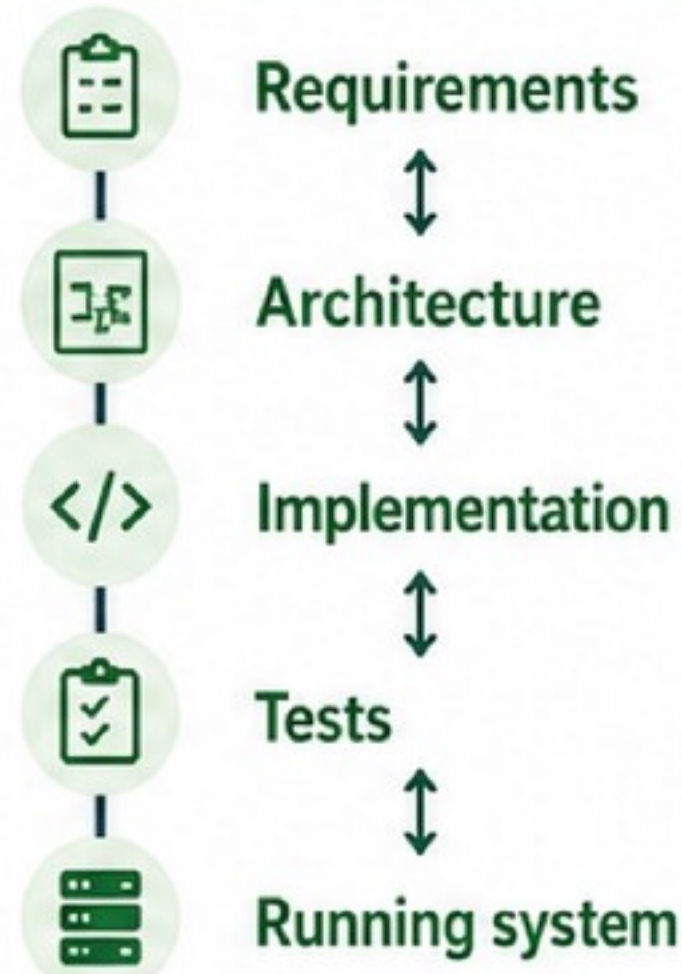
From generated code to a running, verified system



Planned QA: Lifecycle-wide verification



Does the implemented and deployed system satisfy requirements and architectural decisions?



Unit

Integration

Regression

E2E

Performance

Security

Architecture conformance



Planned Deployment:



Build



Package



Containerize



Configure



Deploy



Validate



Code



QA



Deploy



Smoke/E2E



Running system

Docker

Kubernetes

CI/CD

Health checks

Rollback

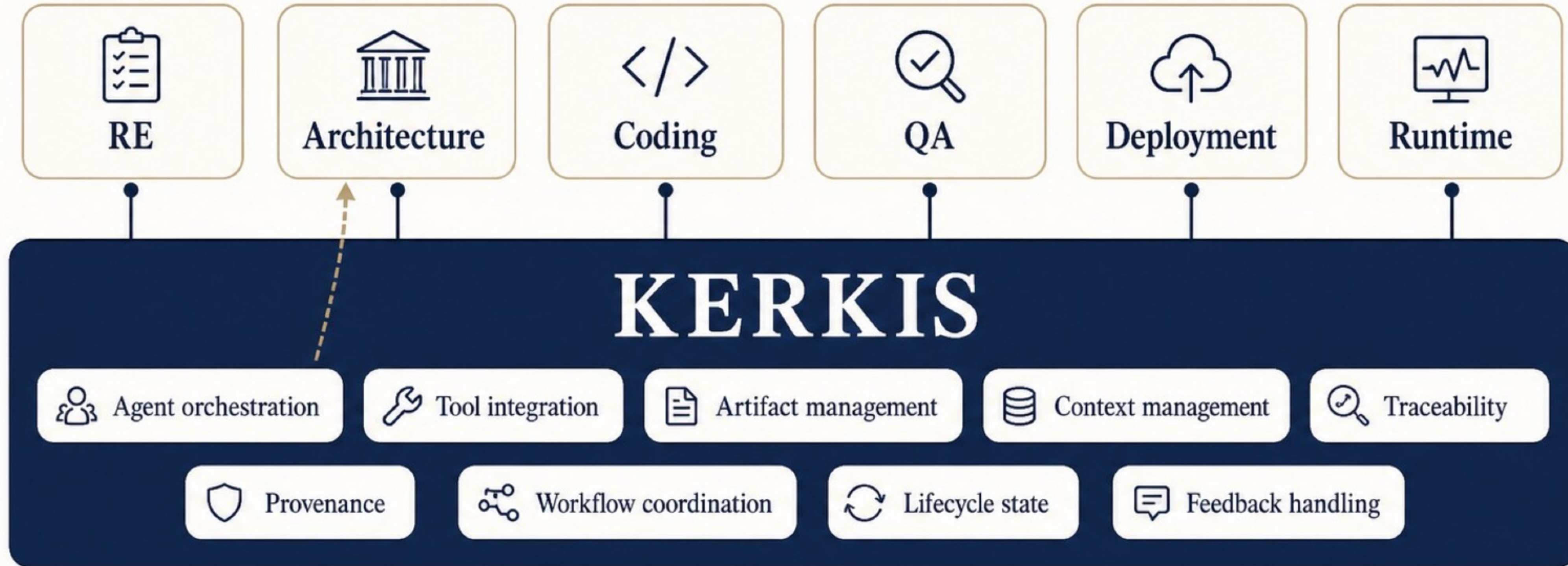


KERKIS

coordinates both planned stages with the rest of the toolchain.

KERKIS — orchestration & integration backbone

Not a lifecycle stage — the layer underneath ANSE



ANSE contains specialized software engineering capabilities.
KERKIS coordinates and integrates those capabilities across the lifecycle.

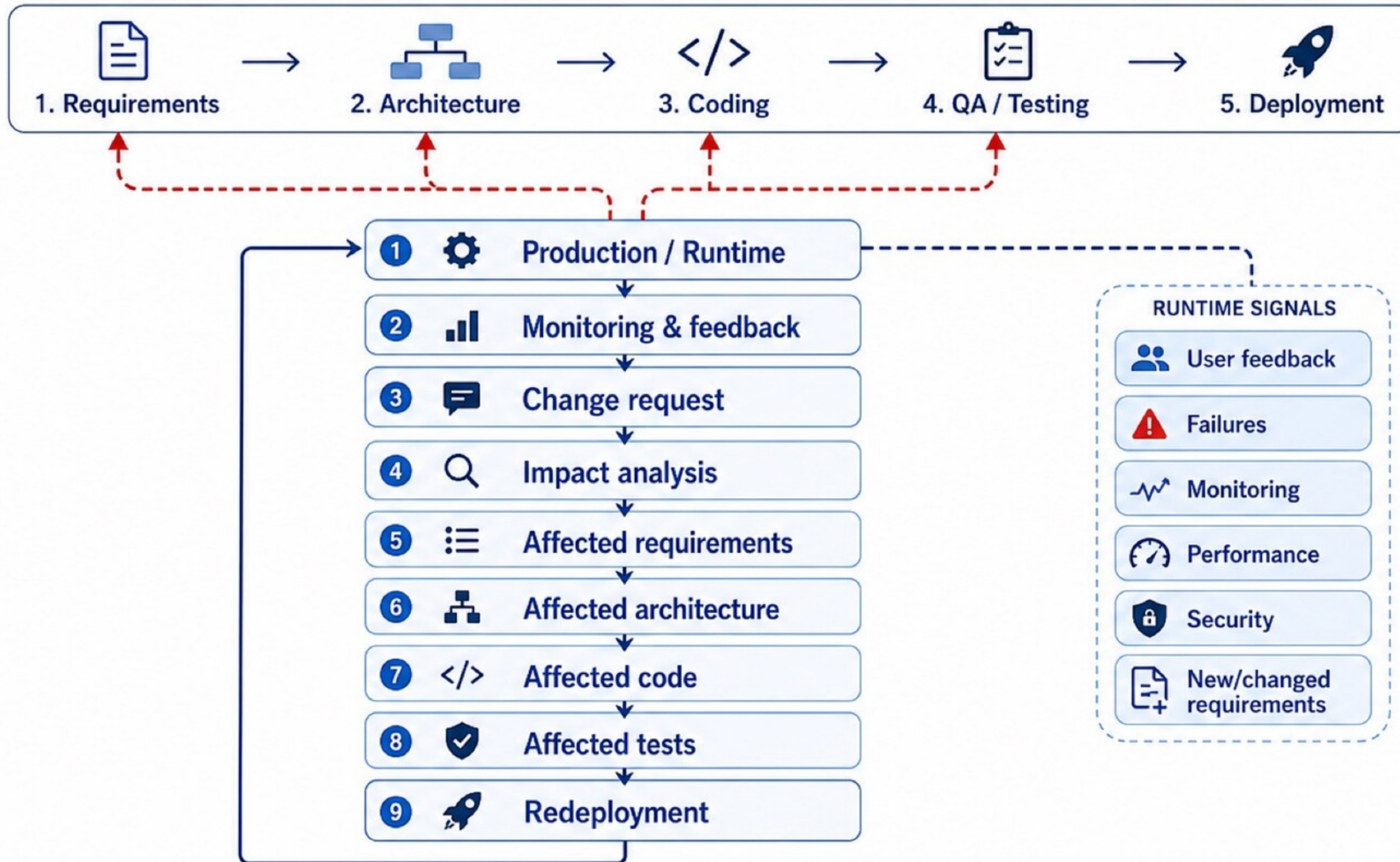


TimeArch is a specialized *Architecture* capability within **ANSE**.*

* TimeArch is one of multiple specialized capabilities that ANSE provides across the lifecycle.

Continuous feedback & change management

ANSE is not a one-shot generator — production closes the loop



Continuous autonomous software evolution.